Roll No. | | | | | | | |

**ODD END SEMESTER EXAMINATION DECEMBER – 2025**

**Name of the course:** B. Tech | **Semester:** 5th
**Name of Paper:** Automotives Fuels & Lubricants | **Paper code:** TME514
**Time:** 3 Hours | **Maximum Marks:** 100

**Note: -**

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

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| **Q1** | (20 marks) | |
| (a) | Explain the classification of fuels based on their physical state with suitable examples. | CO1, CO2 |
| (b) | Discuss the important properties of gasoline and diesel fuels affecting engine performance. | |
| (c) | What are fuel additives? Discuss their types and functions in improving fuel performance. | |
| **Q2** | (20 marks) | |
| (a) | Compare the properties and performance characteristics of gasoline and diesel fuels | |
| (b) | Explain the production, properties, and application of ethanol and methanol as automotive fuels. | CO2, CO3 |
| (c) | Compare the properties, storage, and safety aspects of LPG, CNG, and Hydrogen fuels. | |
| **Q3** | (20 marks) | |
| (a) | A diesel engine consumes 0.25 kg/min of fuel with a calorific value of 43,000 kJ/kg and produces 8.5 kW power. Calculate the brake thermal efficiency of the engine. | CO3, CO4 |
| (b) | What are the functions of automotive lubricants? Classify them with examples. | |
| (c) | Define viscosity index, flash point, and pour point of lubricating oils. | |
| **Q4** | (20 marks) | |
| (a) | A lubricant has viscosities of 120 cSt at 40°C and 12 cSt at 100°C. Determine its viscosity index (VI) using standard ASTM method approximation if the reference oils A and B have viscosities of 14 and 10 cSt respectively at 100°C | C04, CO5 |
| (b) | Derive the stoichiometric air-fuel ratio for the combustion of octane ($C_8H_{18}$). | |
| (c) | Explain the formation of major pollutants such as CO, NOx, and unburnt hydrocarbons. | |
| **Q5** | (20 marks) | |
| (a) | Explain the basic architecture and components of an electric vehicle (EV). | C05 CO6 |
| (b) | Discuss the advantages and limitations of solid-state batteries for electric vehicles. | |
| (c) | Discuss the future prospects of hydrogen and renewable fuels for sustainable transportation. | |